



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

॥वसुधैव कुटुम्बकम्॥

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Course: Data Compression

Course Code: 705210502

LZW Coding
Lempel–Ziv–Welch

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1. Full form: **Lempel–Ziv–Welch**.
2. Type: Lossless compression algorithm.
3. Method: Uses a dictionary-based approach to replace repeated patterns with codes.
4. Initialization: Starts with all single-character entries in the dictionary.
5. Working: Builds new dictionary entries dynamically as sequences appear.
6. Output: Encodes sequences with shorter integer codes.
7. Advantage: Efficient for text with many repeated patterns.
8. Applications: Used in **GIF, TIFF, PDF, Unix compress**.
9. Benefit: Simple, fast, and works well without **prior knowledge of data statistics**.

Encode: wabbab'wqbbab'wabbab'wabbab'w

w a b b a b w a b b a b w a b b a b w

5, 2, 3, 3, 2, 1, 6, 8, 10, 12, 9, 11, 7, 16

LZW Encoding Steps

Initial Given Dictionary

Index	Entry
1	b'
2	a
3	b
4	o
5	w

Index	Entry	Index	Entry
1	b'	11	b'w
2	a	12	wab
3	b	13	bba
4	o	14	ab'w
5	w	15	wabb
6	wa	16	bab'
7	ab	17	b'wa
8	bb	18	abb
9	ba	19	bab'w
10	ab'		

1. Start with a dictionary of all single characters.
2. Set the first input symbol as current string w.
3. Read next symbol a.
4. If w+a is in dictionary → set w = w+a.
5. Else → output code for w, add w+a to dictionary, set w = a.
6. Repeat until input ends.
7. Output code for the final w.

Encoded Code:

“5, 2, 3, 3, 2, 1, 6, 8, 10, 12, 9, 11, 7, 16”



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Thank You